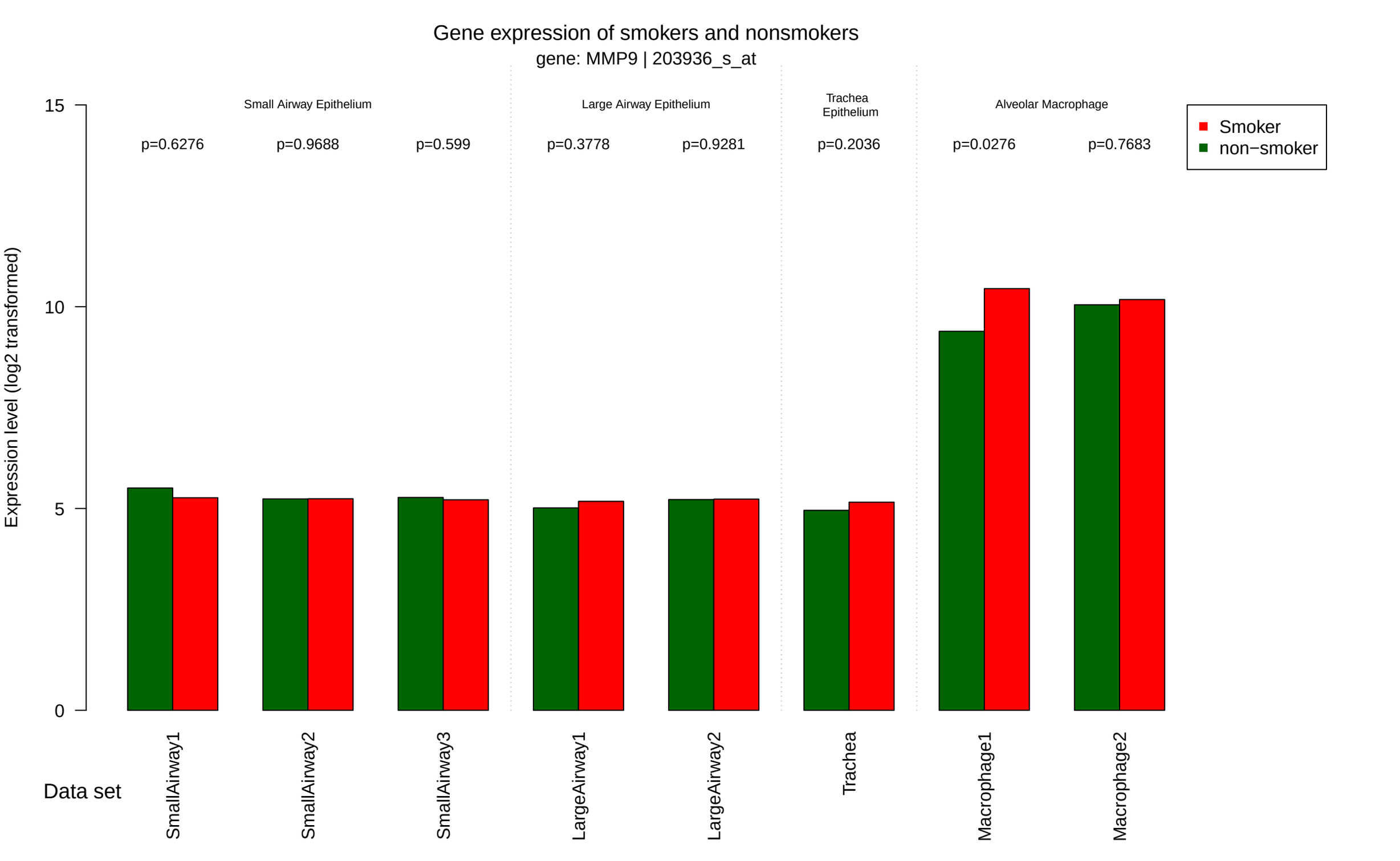
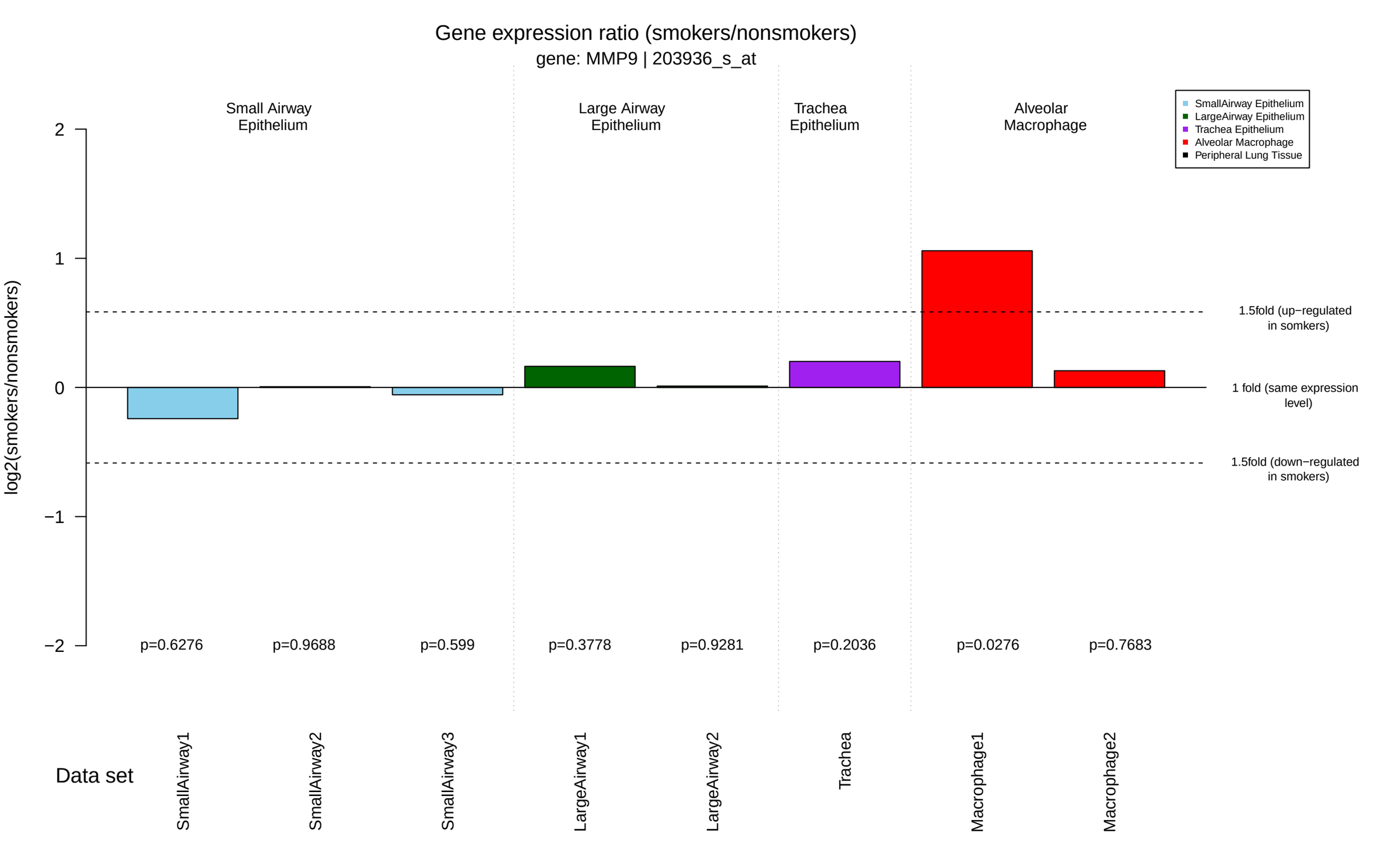
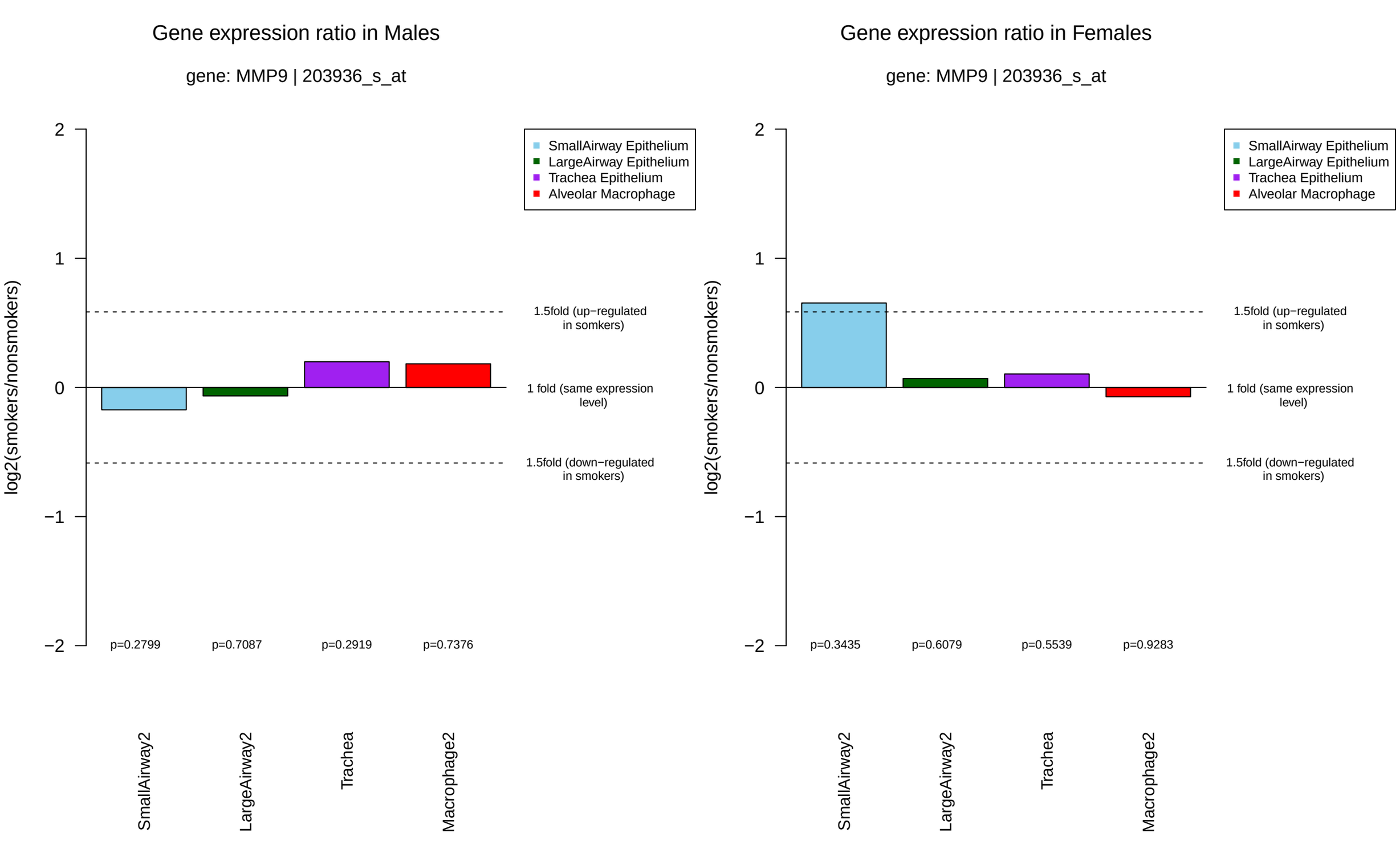
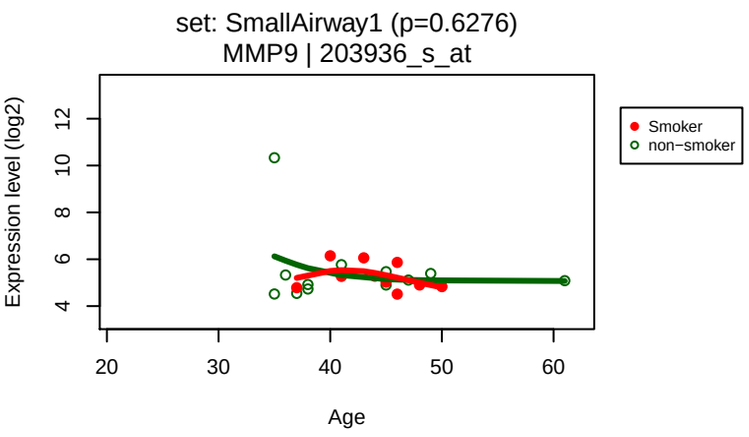
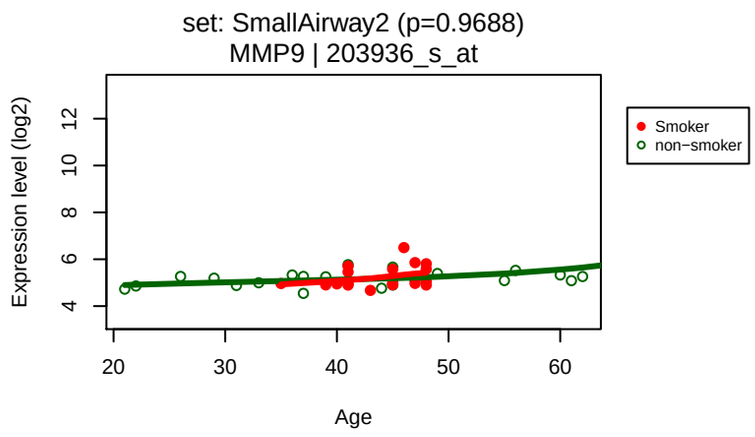
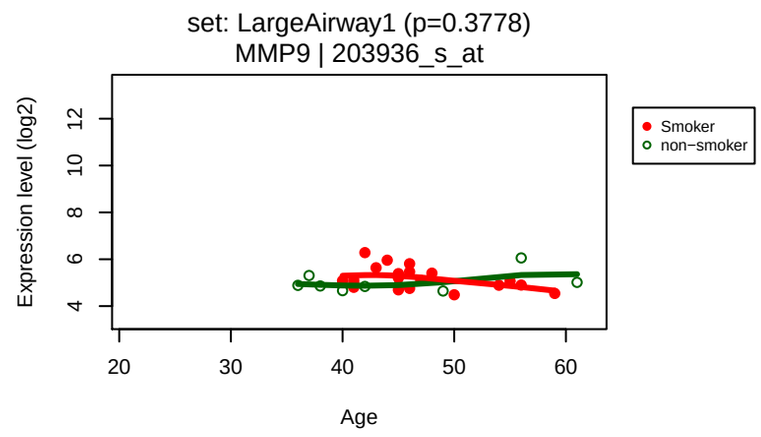

Gene expression of smokers and nonsmokers
gene: MMP9 | 203936_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6276
p=0.9688
p=0.599
p=0.3778
p=0.9281
p=0.2036
p=0.0276
p=0.7683
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

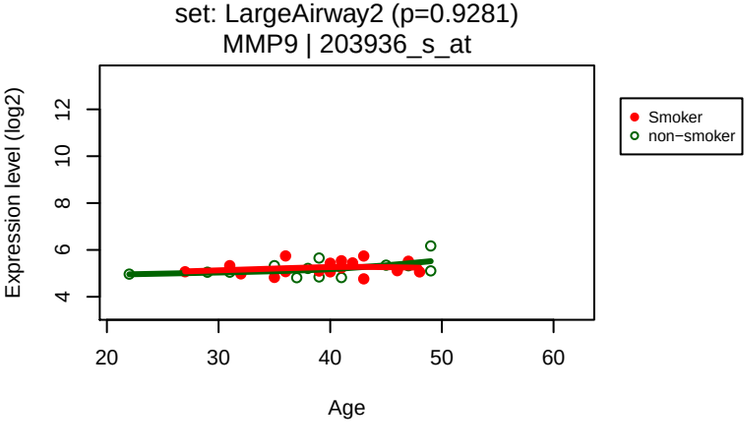

Gene expression ratio (smokers/nonsmokers)
gene: MMP9 | 203936_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6276
p=0.9688
p=0.599
p=0.3778
p=0.9281
p=0.2036
p=0.0276
p=0.7683
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

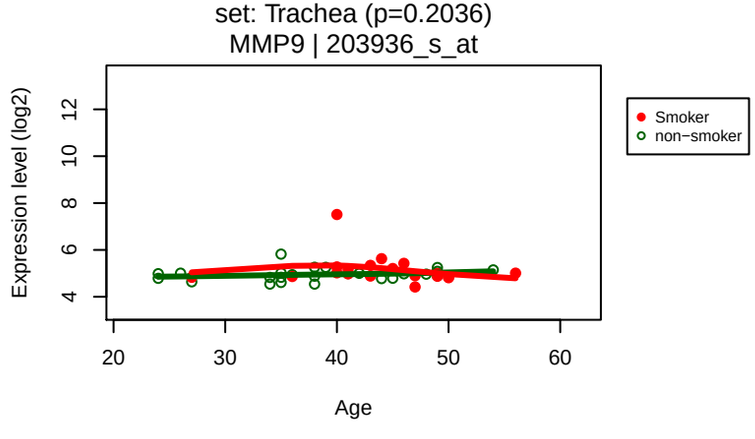

Gene expression ratio in Males
gene: MMP9 | 203936_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2799
p=0.7087
p=0.2919
p=0.7376
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP9 | 203936_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3435
p=0.6079
p=0.5539
p=0.9283
SmallAirway2
LargeAirway2
Trachea
Macrophage2

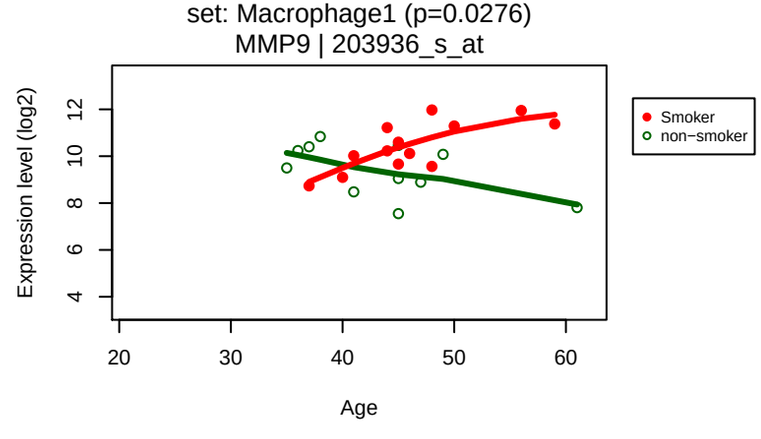

set: SmallAirway1 (p=0.6276)
MMP9 | 203936_s_at
Expression level (log2)
Age
Smoker
non−smoker

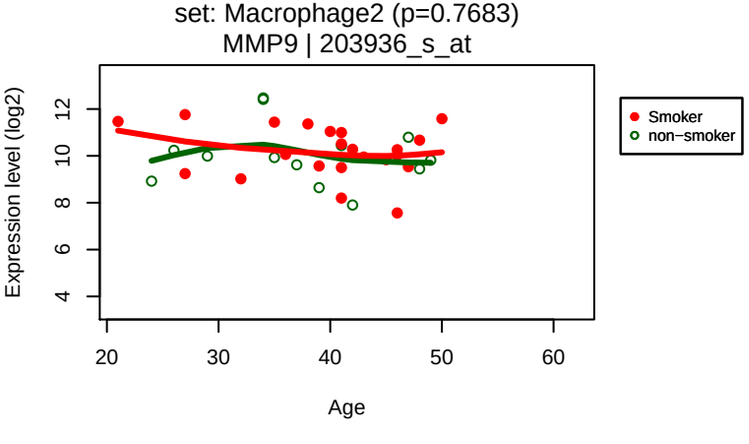

set: SmallAirway2 (p=0.9688)
MMP9 | 203936_s_at
Expression level (log2)
Age
Smoker
non−smoker

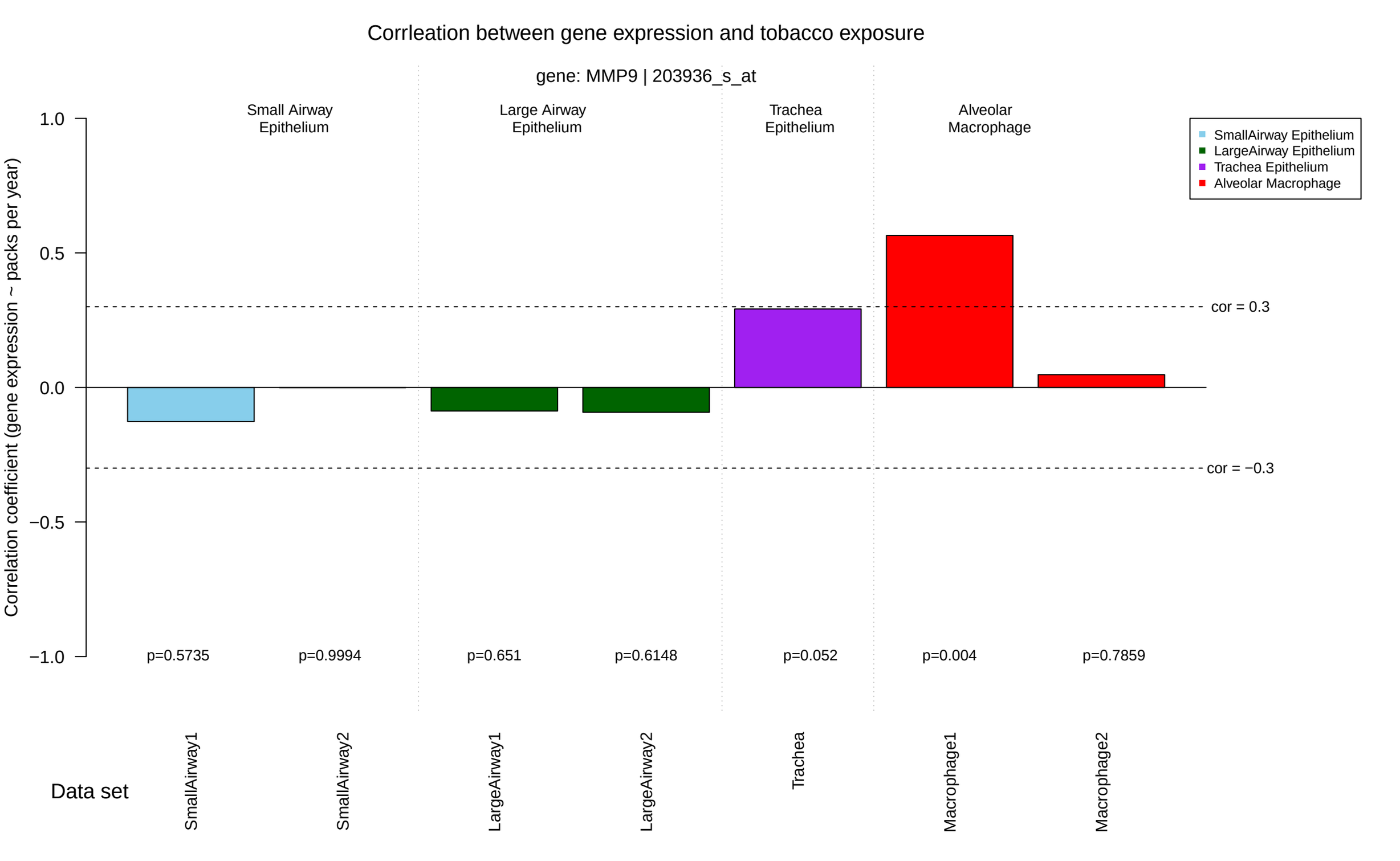

set: LargeAirway1 (p=0.3778)
MMP9 | 203936_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9281)
MMP9 | 203936_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2036)
MMP9 | 203936_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0276)
MMP9 | 203936_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7683)
MMP9 | 203936_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP9 | 203936_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5735
p=0.9994
p=0.651
p=0.6148
p=0.052
p=0.004
p=0.7859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2